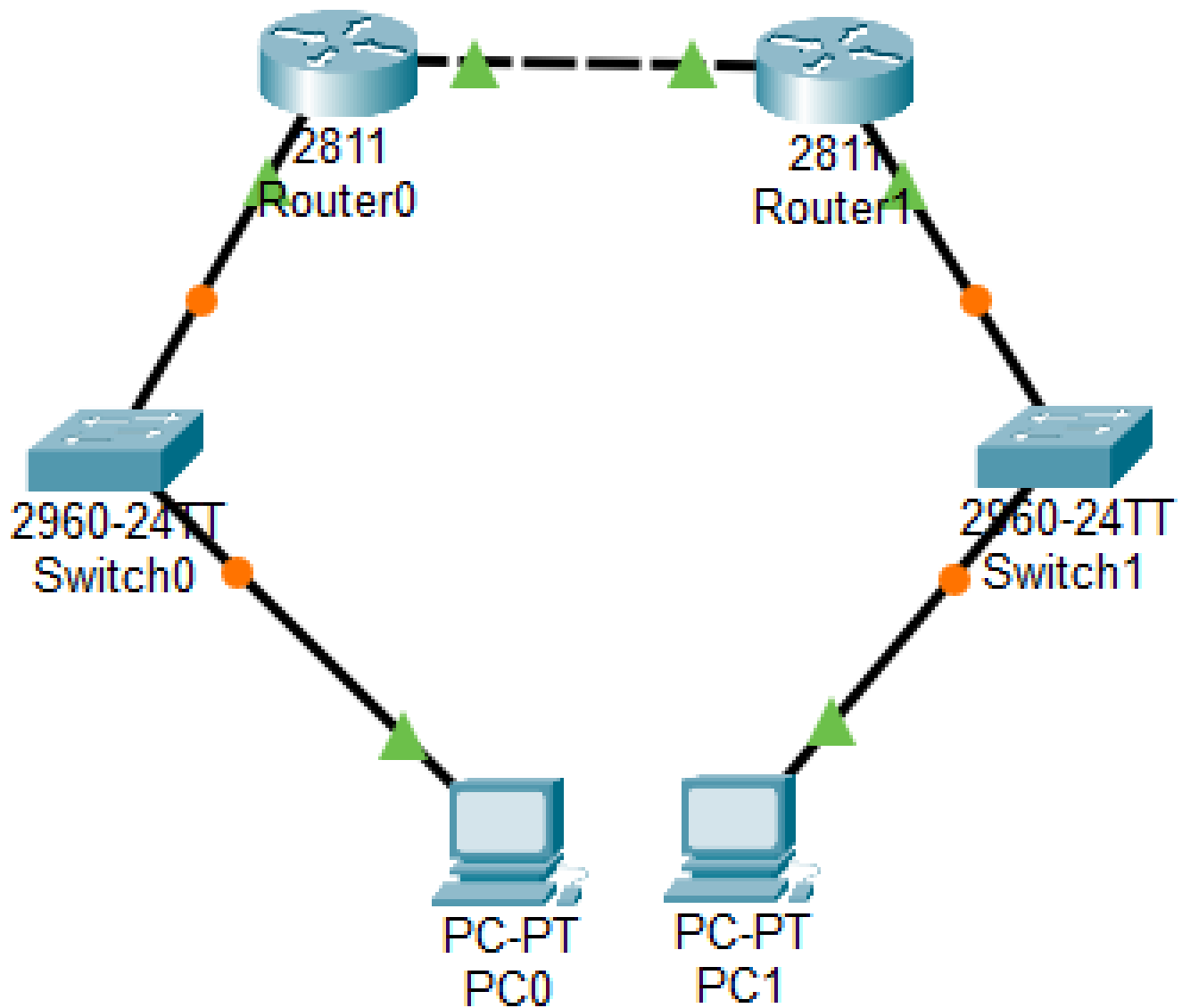


Assignment 5:

Objective:

- Connect **two different networks**
- Configure **static routing manually**
- Enable communication between PCs in different networks

NETWORK:



ROUTER 1 CONFIGURATION:

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int G0/0/0
Router(config-if)#ip add 192.168.1.1 255.255.255.0
Router(config-if)#no shut
Router(config-if)#int Gig0/0/1
Router(config-if)#ip add 10.0.0.1 255.255.255.252
Router(config-if)#no shut
Router(config)#ip route 172.16.0.0 255.255.0.0 10.0.0.2
Router(config)#ex
Router#
%SYS-5-CONFIG_I: Configured from console by console

```

show ip route

```

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

```

Gateway of last resort is not set

```

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0/1
S       172.16.0.0/16 [1/0] via 10.0.0.2
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0/0

```

```

Router#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0/0	192.168.1.1	YES	manual	up	up
GigabitEthernet0/0/1	10.0.0.1	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

ROUTER 2 Configuration:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int Gig0/0/1
Router(config-if)#ip add 10.0.0.2 255.255.255.252
Router(config-if)#no shut

Router(config-if)#ex
Router(config)#int Gig0/0/0
Router(config-if)#ip add 172.16.0.1 255.255.0.0
Router(config-if)#no shut

Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#ex
-
```

VERIFICATION BY ping:

```
C:\>ping 192.168.1.100

Pinging 192.168.1.100 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.1.100: bytes=32 time<1ms TTL=126
Reply from 192.168.1.100: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.100:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.100

Pinging 192.168.1.100 with 32 bytes of data:

Reply from 192.168.1.100: bytes=32 time<1ms TTL=126
Reply from 192.168.1.100: bytes=32 time<1ms TTL=126
Reply from 192.168.1.100: bytes=32 time<1ms TTL=126
Reply from 192.168.1.100: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 172.16.0.100
```

```
Pinging 172.16.0.100 with 32 bytes of data:
```

```
Reply from 172.16.0.100: bytes=32 time=12ms TTL=128
```

```
Reply from 172.16.0.100: bytes=32 time<1ms TTL=128
```

```
Reply from 172.16.0.100: bytes=32 time=6ms TTL=128
```

```
Reply from 172.16.0.100: bytes=32 time=7ms TTL=128
```

```
Ping statistics for 172.16.0.100:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 12ms, Average = 6ms
```